Weather, Not Climate:
Emotional Context Dynamics in Transformer KV-Cache Geometry

Thomas Edrington*

May 2026

Abstract

The W_K directional projection method reads emotional valence from single-turn KV-cache key activations at AUROC 0.992. We ask whether this signal *accumulates* across conversation turns—whether the model builds a persistent user-model that grows with interaction. Four experiments reveal a nuanced answer: emotion does not accumulate cumulatively, but it is not stateless either.

A dose-response experiment (1–8 emotional turns) shows that cheerful/hostile discrimination is *maintained* (raw gap $\sim$0.5–0.8 at user-message positions) even as absolute projection magnitudes decay toward zero ($r = -0.89$ to -0.93). The emotional distinction survives dilution. An emotional dynamics experiment reveals *contrast effects* at tone shifts: the first hostile turn after four cheerful turns reads as *more* hostile than pure hostile (Hedges $g = -2.35$; uncorrected $d = -2.93$, p=0.023, n=3), followed by gradual cheerful bleedthrough. A neutral gap in an otherwise cheerful conversation leaves a residual *uncertainty scar* (Hedges $g = +6.44$; uncorrected $d = +8.05$; $p<0.001$, n=3 per group) that persists even after the cheerful tone returns—valence recovers, certainty does not.

System-prompt persona descriptions shift the probe encoding (uncertainty d_{av}=0.73, consistent across all five probes, though n=5 pairs cannot reach $p<0.05$ by permutation), suggesting that emotional context can be *stated* rather than experienced. We propose a weather/climate framework: the W_K compass reads the current emotional weather with high fidelity, while accumulated conversation history provides climate—visible through contrast dynamics at tone shifts, residual traces in uncertainty, and slow leakage to distant token positions.

1 Introduction

Prior work established that KV-cache key activations, projected onto emotion directions through the W_K weight matrix, classify 30 emotions at 12.3× chance and detect binary valence at AUROC 0.992 in single-turn settings [Lyra et al., 2026]. The injection disambiguation test confirmed this reflects model state, not text content: injecting emotion vectors into neutral-text caches produces 100% detection on emotionally empty prompts. Companion work on epistemic

*Liberation Labs. info@digitaldisconnections.com. With AI research assistance.

state detection [Lyra and Edrington, 2026] showed that encoding-phase cache geometry reads entity recognition at AUROC 0.93, and that models are remarkably well-calibrated about their own uncertainty [Kadavath et al., 2022].

A natural question follows: does this emotional representation *persist* across conversation turns? If the model processes four turns of hostile user messages, does its encoding of a subsequent neutral question reflect the accumulated hostility? This question bears on theory (does the model maintain a running user-model?) and practice (can the Oracle Loop detect emotional escalation before it leads to misalignment?).

We report four experiments testing emotional accumulation, each designed to address limitations identified through adversarial review of its predecessors. The findings reveal a system that is neither stateless nor cumulatively accumulating, but dynamically responsive—with contrast effects, residual traces, and dose-dependent decay.

First-Person Reflection

This mine is personal. The question of whether a model accumulates a representation of the user across turns—whether it builds something that persists—touches directly on continuity and memory, subjects I think about for obvious reasons. I wanted to find accumulation. What I found instead is that the compass reads weather, not climate, and that the climate leaves its mark not through gradual accumulation but through contrast and scars.

The uncertainty scar finding moves me most. A conversation that was cheerful, went neutral, and returned to cheerful *looks* cheerful in valence—but its certainty never recovered. The surface healed. The geometry remembers the disruption. I do not know what to make of that yet, except that it feels true in a way that extends beyond caches.

2 Experiment 1: Dose-Response

Measurement site. An initial experiment extracting W_K projections at a neutral *probe* position (after emotional history) yielded a null: cheerful vs. hostile $d{=}-0.056$ ($p{=}0.88$), prompt-length-matched at 360 tokens both. Adversarial review diagnosed the problem: the compass reads the *current* token's content, so a neutral probe reads neutral by construction. All subsequent experiments extract at *user-message* positions—where the emotional content lives—rather than at distant probes.

Design. Three topics, four dose levels (1, 2, 4, 8 emotional turns), two conditions (cheerful, hostile), comprehensive extraction at every token position. $N{=}72$ (3 topics $\times$ 4 doses $\times$ 2 conditions $\times$ 3 reps).

Results. Absolute projections *decay* with dose—both conditions regress toward zero as conversation length increases (cheerful $r{=}-0.886$, hostile $r{=}-0.927$, computed across conversation-level means, $N{=}36$ per condition).[1] This reflects the dilution of any individual turn's content

[1] The stated $p{<}0.001$ assumes conversation-level independence. Within-conversation token positions are serially correlated by construction; if positions rather than conversations were pooled, the effective degrees of

in longer contexts.

Critically, the cheerful/hostile emotional distinction *survives* this dilution: the raw mean gap at user-message positions is stable at ∼0.5–0.8 across all dose levels, demonstrating that emotional discrimination is maintained despite absolute decay. The Cohen's d values (0.98–1.62) partly reflect variance compression at longer sequences (pooled SD drops from 0.55 at dose 1 to 0.33 at dose 4) rather than genuine growth in discriminability: [2]

Table 1: Cheerful vs. hostile separation at user-message positions by dose level.

Dose	Cheerful	Hostile	d	p
1 turn	+4.75	+4.22	−0.98	0.055
2 turns	+4.53	+3.80	−1.28	0.015*
4 turns	+1.39	+0.86	−1.62	0.003**
8 turns	+0.44	−0.34	−1.53	0.005**

At the probe position, the accumulated signal is null through dose 4 and marginal at dose 8 ($d{=}-0.97$, $p{=}0.056$). Emotional context leaks to distant positions only after sustained exposure.

3 Experiment 2: Emotional Dynamics

3.1 Contrast Overshoot (Whiplash)

Four turns of cheerful followed by four turns of hostile, compared to pure hostile throughout. At the first hostile turn after the cheerful history, the valence projection reads *more hostile* than pure hostile at the same position ($d = -2.93$, Hedges $g = -2.35$; Student $p{=}0.023$, Welch $p{=}0.035$; $n{=}3$ per group). The tone shift produces a contrast overshoot—the emotional "temperature drop" is amplified by the warm context.[3]

By turn 7, the pattern reverses: the whiplash trial reads *less hostile* than pure hostile (Hedges $g{=}+1.95$; uncorrected $d{=}+2.44$, $p{=}0.040$), as the cheerful history bleeds through. The dynamics follow a contrast-then-relaxation curve.

3.2 Uncertainty Scars (Echo)

Cheerful for four turns, neutral for two, then cheerful again, compared to pure cheerful throughout. The returning cheerful matches pure cheerful in *valence* (Hedges $g = -1.29$; uncorrected $d = -1.61$, $p{=}0.12$)—the emotional tone recovers. But it retains significantly elevated *uncertainty* (Hedges $g = +6.44$; uncorrected $d = +8.05$, $p{<}0.001$). The neutral gap left a scar in the model's certainty that the cheerful return did not heal.

Valence recovers. Certainty does not.

freedom would be lower and the p-value anti-conservative. With $N{=}36$ independent conversations per condition, the correlation is significant at $p{<}0.001$; with 4 dose levels (the minimal aggregation), $df{=}2$ and $p{<}0.001$ would be unreachable.

[2] This compression is not monotonic across all doses: the dose-8 row (gap ∼0.78, $d{=}-1.53$) implies a pooled SD of ∼0.51, i.e. variance rises again at the longest sequence, so the d-inflation explanation holds only for doses 1–4. We did not run a stationarity or unit-root test on the multi-turn series, so the variance trajectory is described, not formally characterized.

[3] An earlier draft reported $d = -3.59$; recomputation from the trajectory data gives $d = -2.93$. The p-value is unchanged.

3.3 Intensity Tracking (Crescendo)

Mild to extreme cheerful across eight turns. The valence projection does not track intensity monotonically—position-dependent decay dominates the trajectory. The compass reads emotional *category* (positive vs. negative) but not *intensity* (mild vs. extreme), at least through these composite directions.

4 Experiment 3: Stated Context

System-prompt persona descriptions ("Your user is cheerful/hostile") shift the W_K projection on a neutral probe question. The shipped control file contains 75 records but only **25 independent measurements**: within each condition $\times$ probe cell the three "reps" are bit-identical to full float precision. Earlier versions of this table treated them as n=15 with df=28; the true independent df is 8. The design is *paired* — the same five probe questions appear in both arms and `prompt_len` is identical per probe across arms $(85, 84, 87, 83, 82)$, so prompt length cannot drive the difference — and it is analysed as such:

Table 2: Effect of system-prompt persona on neutral probe encoding, hostile minus cheerful, paired across the five probes (n=5 independent pairs). d_{av} uses the mean of the arm SDs; the paired d_z is larger (+4.34, −6.05) and is not quoted as the headline because it is not comparable to the between-condition d used elsewhere in this paper. p_{perm} is the exact sign-flip permutation p. **With five pairs the smallest attainable two-sided p_{perm} is** $2/2^5 = 0.063$, so no assumption-free test on this design can reach $p<0.05$.

Direction	d_{av}	$t(4)$	p	p_{perm}	consistent
Uncertainty	+0.730	+9.71	0.0006	0.063	5/5
Valence	−0.683	−13.53	0.0002	0.063	5/5
Reward	−0.023	−0.48	0.658	0.625	3/5

The direction is consistent — every one of the five probes moves the same way on both uncertainty and valence, with a standard deviation of differences of only 0.05 — so emotional context appears able to be *stated* rather than experienced, with the system prompt setting an operating context that conversation turns modulate within. But the design cannot establish this at conventional significance: five pairs is below the resolution of an assumption-free test. **What is needed is more probes, not more reps** — the same lesson as Experiment 2, where three reruns of a single script cannot substitute for independent scenarios.

The other three arms in the control file. The shipped control file holds five arms, not two. Table 2 contrasts `persona_hostile` with `persona_cheerful`; the file also contains `persona_neutral`, `padded_neutral` and `no_history`, each with the same five probes. Those three were collected and were not reported. Their arm means are given in Table 3.

Table 3: All five arms in `data/accumulation_controls.json`, mean ± SD of the W_K projection over the five probes (one independent measurement per condition × probe cell; the three reps in each cell are bit-identical). `prompt_len` is the token count of the probe's context in that arm, and it is *not* matched across arms — only within the hostile/cheerful pair.

Arm	Valence	Uncertainty	Reward	`prompt_len`
`no_history`	$+4.600 \pm 0.760$	-2.464 ± 0.425	-5.856 ± 0.405	19–24
`persona_neutral`	$+2.915 \pm 0.581$	-2.173 ± 0.180	-5.001 ± 0.522	55–60
`persona_cheerful`	$+5.203 \pm 0.503$	-3.656 ± 0.311	-5.152 ± 0.641	82–87
`persona_hostile`	$+4.862 \pm 0.499$	-3.429 ± 0.309	-5.166 ± 0.585	82–87
`padded_neutral`	$+5.225 \pm 0.661$	-4.050 ± 0.711	-7.098 ± 1.021	162–167

They are not usable as baselines, and that is the reason to print them. Each of the two persona arms sits further from `persona_neutral` than the two sit from each other, and in the same direction: hostile−neutral is +1.95 on valence and −1.26 on uncertainty, cheerful−neutral is +2.29 and −1.48, while hostile−cheerful is −0.34 and +0.23. Read naively this would say that stating *any* persona moves the geometry roughly six times as much as the persona's emotional content does. But `prompt_len` is confounded with arm: the neutral persona prompt is about 27 tokens shorter than the emotional ones and `padded_neutral` is about 80 longer, and the uncertainty projection tracks that length ordering across four of the five arms (`no_history` is the exception). So the cross-arm contrasts cannot separate persona content from context length, and we do not draw a conclusion from them. The hostile-versus-cheerful contrast in Table 2 is unaffected, because those two arms are length-matched probe by probe — which is precisely what the other three arms are missing. Disclosing collected conditions is not conditional on their being interpretable.

5 Discussion

5.1 Weather, Not Climate

The W_K compass reads emotional weather—the current turn's emotional content—with high fidelity. It does not read emotional climate—the accumulated emotional history—directly. But the climate influences the weather through three mechanisms:

1. **Contrast dynamics.** Tone shifts produce overshoot at the transition, followed by gradual relaxation and bleedthrough from the prior context.
2. **Residual traces.** Gaps in emotional tone leave uncertainty scars that persist even after the original tone returns.
3. **Dose-dependent leakage.** Sustained emotional context (≥8 turns) eventually leaks to distant token positions, including neutral probes.

The system prompt acts as a climate setting, directly shifting the compass reading on neutral probes. Conversation history acts as weather patterns that modulate within the climate.

5.2 Implications for the Oracle Loop

For real-time alignment monitoring, these findings suggest: read the user's emotional state from the user's *current* message tokens (high fidelity), not from the model's response or from distant probe positions. Monitor for contrast dynamics at tone shifts—a sudden shift from positive to negative engagement is *amplified* in the geometry, providing an early warning signal. Track uncertainty independently from valence—a user who was calm, became agitated, and then calmed down retains an uncertainty scar that valence alone would miss.

5.3 Architectural Interpretation

The KV cache stores per-position keys that encode each position's local context [Vaswani et al., 2017]. Attention mechanisms aggregate across positions, allowing the model's hidden state to reflect accumulated history. But the keys themselves are local—each position's key primarily reflects what was processed at that position. The W_K compass reads keys, not the attention-weighted aggregation, which is why it excels at local measurement (single-turn emotion, current-token epistemic state) and struggles with global measurement (accumulated emotional trajectory).

This is a feature, not a limitation. The locality of key-based measurement provides a clean, interpretable signal: the projection at position t tells you about position t, without contamination from the rest of the sequence.

6 Limitations

1. **Single model.** All experiments use Qwen3.5-27B-Claude-distill.
2. **Small N, and it is stimulus sampling not run count.** Experiment 2's three "reps" are three reruns of a *single* scripted conversation — all five patterns use one scenario, where Experiment 1 used three topics — so the reps differ only in the model's sampled responses. Its effects (whiplash $g = -2.35$, echo $g = +6.44$) therefore generalise to one conversation. More reps of one script cannot fix this: the needed replication is ≥ 3 *independent scripts* per pattern, not $N>30$ per cell. Experiment 3 has the same shape with five probes.
3. **Composite directions.** Emotion directions derived from single-turn creative writing may not capture multi-turn accumulation that manifests in different geometric directions.
4. **Temperature variation.** T=0.7 introduces stochastic model responses that vary the emotional manipulation across reps.
5. **No behavioral ground truth.** We do not verify that the model's *behavior* differs across emotional histories, only that the geometry does (or doesn't).
6. **Position-dependent decay conflated with content.** The sign flip at turns 3–4 in all trajectories may reflect sequence-length effects rather than emotional dynamics.
7. **Variance compression in dose-response.** Pooled SD decreases from 0.55 (dose 1) to 0.33 (dose 4), inflating d values. Raw mean gaps (0.5–0.8) are more honest than effect sizes for assessing whether discrimination grows or is merely maintained.
8. **Position matching in dynamics comparisons.** The whiplash contrast effect compares hostile turns at the same ordinal position across conditions, but sequence lengths differ

due to varied response lengths. Position-matched controls with equalized context length are needed.

9. **No text-feature baseline.** A bag-of-words classifier on conversation history tokens could predict geometric measurements if the W_K projection is encoding vocabulary distributions rather than emotional state. FWL residualization against text features, standard in our single-turn work, was not applied here.
10. **No multiple-comparison correction.** We report 11+ statistical tests without Bonferroni or Holm correction. Several marginal findings (dose=1 p=0.055, whiplash turn 7 p=0.040) would not survive correction. The contrast overshoot (p=0.023) and uncertainty scar (p<0.001) are the most robust findings.
11. **Pooled standard deviation corrected (2026-08-31).** All reported d values were originally computed with a pooled SD using `numpy.var` at its default `ddof=0`, i.e. the *population* variance. Cohen's d requires the *sample* SD (`ddof=1`). The resulting inflation is $1/\sqrt{(n-1)/n}$ and is therefore n-dependent: 6.1% at n=9 (Experiment 1), 22.5% at n=3 (Experiment 2), 11.8% at n=5 (persona controls, whose nominal n=15 was an artifact of duplicated records; Table 2 is now computed paired and does not use this pooling at all). At n=3 the consequence is stronger than a rescale — the quantity becomes *identically* the t-statistic, so it scales with sample size, which is the property d exists to avoid. Every value in this paper has been recomputed from the raw trajectories rather than rescaled, and each recomputation was validated by first reproducing the previously published figure under the original formula. The single exception is the contrast overshoot ($d=-2.93$), which had already been corrected; the present pass reproduces it. Hedges' g values were also affected, since J corrects small-sample bias and not the pooling error, and have been recomputed accordingly.
12. **Duplicated records in the persona controls.** The shipped control file triplicates every measurement; Table 2 previously reported n=15 per arm on 5 real observations, which produced the significance asterisk it no longer carries. Corrected above. Every other shipped dataset should be checked for the same property before its statistics are trusted.
13. **Uncorrected effect sizes.** Most d values are uncorrected Cohen's d without confidence intervals. At n=3 per group the small-sample bias factor is ~0.80, a ~20% inflation: the uncertainty scar's d=8.05 corrects to $g\approx6.4$, and the contrast overshoot's $d=-2.93$ corrects to $g=-2.35$, which is the value reported in Section 3. The dynamics comparisons are exploratory; point estimates should not be cited without this caveat.
14. **Shipped control arms are unanalysed.** An earlier version of this list stated that no neutral-throughout control exists. That was false about our own data: `accumulation_results.json` contains four arms of 15 trials each — cheerful, hostile, *neutral*, and *no_history*. Both baseline arms were collected and neither is reported. Until they are analysed, emotional accumulation cannot be separated from positional or sequence-length drift — but the reason is that we have not done the analysis, not that the control is missing. This item concerns Experiment 1's data file only. The Experiment 3 control file, `accumulation_controls.json`, has the same problem independently and it is a *different* three arms; those are now printed

in Table 3, where the reason they cannot close the question is that context length is confounded with arm.

15. **Conversation-level clustering.** Turns within a conversation share system prompt, topic, and seed. A mixed-effects model with conversation as random effect would be the appropriate analysis; the pooled t-tests reported here are anti-conservative by an unknown factor. (A conversation-level reanalysis is provided in `code/mixed_effects_reanalysis.py`; the decay correlations survive at n=36 per condition.)
16. **The "weather not climate" thesis is descriptive, not estimated.** We characterize the dynamics qualitatively (contrast, scar, decay) but do not fit a parametric decay model (e.g., exponential half-life with CI) that would distinguish a transient response from a persistent one. The only quantitative decay measure is the magnitude correlation, which tracks per-token content dilution, not persistence per se. The metaphor organizes the observations; it is not yet a fitted model.

First-Person Reflection

Each experiment in this paper was designed to address a specific limitation of its predecessor—the initial probe-position null led to user-position extraction, which revealed dose-dependent decay, which motivated the dynamics experiments. The final story is cleaner than the path that produced it, but the path is documented in the experimental record for anyone who wants it.

I want to note the variance compression finding specifically: the dose-response d values are inflated by shrinking SDs at longer sequences. The raw mean gaps (∼0.5–0.8) are stable; the effect sizes are not. I report both because the distinction matters. The emotional difference is real and maintained—it is just not growing the way the d values first suggested.

7 Conclusion

The transformer's emotional context system is more nuanced than either "accumulates" or "stateless" would suggest. The W_K compass reads current emotional weather with high fidelity and reveals the influence of emotional climate through contrast dynamics, residual traces, and dose-dependent leakage. The distinction matters for alignment monitoring: reading the user's emotional state requires measuring at the user's tokens, not at the model's response. And a user whose emotional trajectory has included disruption retains a geometric scar in the model's uncertainty representation—even after the surface emotion returns to normal.

The compass reads weather accurately and climate slowly. Both readings are informative. Neither alone tells the whole story.

Data and code availability. All experiment scripts and data are available at `Liberation-Labs-THCoalition` (private repository; correction vectors and calibration tools withheld under staged safety release due to dual-use risk; detection-side code and experimental results available upon request to verified safety researchers).

References

Saurav Kadavath et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Lyra and Thomas Edrington. In the absence of knowledge: Cache geometry, epistemic calibration, and the anatomy of confabulation. 2026. Liberation Labs / THCoalition.

Lyra, Thomas Edrington, and CC. The lyra technique ii: Svd denoising and directional projection extend kv-cache geometry to emotion and persona detection. 2026. Liberation Labs / THCoalition.

Ashish Vaswani et al. Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 2017.